

Transparent, super-stretchy skin-like sensor

Stanford researchers have developed an elastic, transparent skin-like pressure sensor, which can sense pressure from a firm pinch to thousands of kilograms.

The key element of the sensor is a transparent film of carbon 'nanosprings,' which is created by spraying nanotubes in a liquid suspension onto a thin layer of silicone, which is then stretched. The sensor can be stretched in all directions, which then rebounds to the original shape.

The sensors consist of two layers of the nanotube-coated silicone, which are oriented such that the coatings are face-to-face, with a layer of a more easily deformed type of silicone between them. The middle layer of silicone stores electrical charge, much like a battery. When pressure is exerted on the sensor, the middle layer compresses, which alters the amount of electrical charge it can store.

That change is detected by the two films of carbon nanotubes, which act like the positive and negative terminals on a typical automobile or flashlight battery. The change sensed by the nanotube films enables the sensor to transmit what it is 'feeling.'

Solar paint that generates electricity

Imagine the paint on your home's exterior walls generating electricity from sunlight, which can be used to power your appliances and equipment! A team of researchers at the University of Notre Dame, USA, has created a solar paint—Sun-Believable—that uses semiconducting nanoparticles to produce energy.

The team coated nano-sized particles of titanium dioxide with either cadmium sulphide or cadmium selenide. The particles were then suspended in a



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water-alcohol mixture to create a paste. When the paste was brushed onto a transparent conducting material and exposed to light, it created electricity.

"Semiconductor nanocrystals exhibit size-dependent optical and electronic properties. In a quantum dot-sensitized solar cell, the excitation of semiconductor quantum dot or semiconductor nanocrystal is followed by electron injection into titanium dioxide nanoparticles. These electrons are then transferred to the collecting electrode surface to generate photocurrent. The holes that remain in the semiconductor quantum dot are removed by a hole conductor or redox couple and transported to a counter electrode.

Self-healing system to restore broken circuits

What if a cracked chip could fix itself, so fast that the user never knew there was a problem? University of Illinois professors Nancy Sottos, Scott White and Jeffrey Moore have developed a technology that could extend the longevity of electronic devices and batteries.

Earlier, the team had developed a system for self-healing polymer mate-

rials, which they decided to adapt for conductive systems. They dispersed tiny microcapsules, as small as 10 microns in diameter, on top of a gold line functioning as a circuit. As a crack propagates, the microcapsules break open and release the liquid metal contained inside. The liquid metal fills in the gap in the circuit, restoring electrical flow. A failure interrupts current for a few microseconds as the liquid metal starts filling the crack.

Wireless capsule endoscope controlled by MRI

Researchers at Tel Aviv University (TAU), Israel, are developing a capsule endoscope that can move through the digestive tract independent of any attachment.

According to Dr Gabor Kosa of TAU's School of Mechanical Engineering, these wireless capsules will use the magnetic field of a magnetic resonance imaging (MRI) machine and electronic signals manipulated by those operating the capsule to forge a more precise and deliberate path.

In order to help the capsules 'swim' with the magnetic current, the research-